

Shitij Govil

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sgovil.me | [Google Scholar](https://scholar.google.com/citations?user=sgovil5) | [linkedin.com/in/shitij-govil](https://www.linkedin.com/in/shitij-govil) | github.com/sgovil5

EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Science; GPA: 3.95/4.00

Atlanta, GA

Expected December 2026

MANUSCRIPTS UNDER REVIEW / IN PREPARATION

1. **Shitij Govil***, J. A. Collins*, N. Patel*, A. Agarwal*, A. Garg. *FloMo: 3D Scene Flow as a World-Action Model Intermediate for Learning from Human Video*. Submitted to **ICRA**.
 2. S. Miao*, **Shitij Govil***, J. P. Rodgers, M. Liu, J. Duarte, S.-C. Hsu, Y.-T. Chou, P. Li. *HEPTv2: End-to-End Efficient Point Transformer for Charged Particle Reconstruction*. [paper]; under review at *PRX Intelligence*.
 3. Y. Huang, **Shitij Govil**, B. Dai, P. Li. *Self-Rollout Seq-Flow for Efficient Forecasting with Controlled Error Accumulation* (working title). In preparation for submission to **ICLR**.
- * Equal contribution.

SELECTED PUBLICATIONS

1. C. Hashemi, B. Ravichandran, C. Anderson, **Shitij Govil**, B. RichardWebster, B. Hu, K. Fieldhouse. *Improving Semantic Coverage in AI Test and Evaluation through Generative Augmentation*. In *Assurance and Security for AI-enabled Systems 2026*, Proc. SPIE 14046, 1404612 (2026). [paper]
2. **Shitij Govil**, J. P. Rodgers, Y.-T. Chou, S. Miao, A. Saha, A. Anand, K. Lieret, G. DeZoort, M. Liu, J. Duarte, P. Li, S.-C. Hsu. *Locality-Sensitive Hashing-Based Efficient Point Transformer for Charged Particle Reconstruction*. Machine Learning and the Physical Sciences Workshop, **NeurIPS 2025**. [paper] [poster]
3. W. Du, B. Kisananin, G. Armstrong, S. Toshniwal, I. Moshkov, A. Ayrapetyan, S. Mahdavi, D. Zhao, S. Diao, D. Masulovic, A. Avadhanam, M. Wang, **Shitij Govil**, et al. *Is Human-Written Data Enough? The Challenge of Teaching Reasoning to LLMs Without RL or Distillation*. AI for Math Workshop, **ICML 2025**. [paper] [poster]
4. S. Jia, **Shitij Govil**, M. Ramaprasad, V. Fung. *Pre-training Graph Neural Networks with Structural Fingerprints for Materials Discovery*. [paper] (2025).
5. **Shitij Govil**, A. J. Lee, A. C. MacQueen, N. G. Pricope, A. Minei, C. Chen. *Using Hyperspatial LiDAR and Multispectral Imaging to Identify Coastal Wetlands Using Gradient Boosting Methods*. *Remote Sensing*, 14(23), 6002 (2022). [paper]

RESEARCH EXPERIENCE

PAIR Lab – Georgia Institute of Technology

Student Researcher (Advisor: Animesh Garg)

Aug 2025 – Present

Atlanta, GA

- Investigate world-action models that learn robot manipulation from unlabeled human video, using 3D scene flow as an embodiment-agnostic representation of motion and action.
- Scale world-action models with larger, more diverse human and robot video datasets, studying how dataset size and composition affect generalization across embodiments and tasks.

GCOM Lab – Georgia Institute of Technology

May 2024 – Present

*Student Researcher (Advisor: Pan Li)**Atlanta, GA*

- Develop adaptive re-noising methods for efficient generative forecasting, using self-rollout flow models to control error accumulation over long prediction horizons.
- Develop efficient point transformers for charged-particle reconstruction, focusing on locality-sensitive hashing and learned object segmentation for large, irregular detector data.

Fung Lab – Georgia Institute of Technology

Aug 2023 – Jan 2025

*Student Researcher (Advisor: Victor Fung)**Atlanta, GA*

- Investigated graph-neural-network pre-training with structural fingerprints as inexpensive proxy supervision for materials-property prediction.

INDUSTRY EXPERIENCE**Amazon**

May 2026 – Aug 2026

*Software Engineering Intern**Los Angeles, CA*

- Developed a chatbot for satellite-management tools for Amazon LEO.

Kitware

May 2025 – Aug 2025

*Machine Learning Engineer Intern**Minneapolis, MN*

- Proposed and led implementation of a diffusion-based data-augmentation module, designing its core API, plugin system, and data-flow pipeline for an open-source library.

AWARDS AND SERVICE**Awards:** Presidential Undergraduate Research Award and PURA Travel Award, Fall 2025**Service:** Conference on Robot Learning (CoRL), 2026; ICML AI for Math Workshop, 2025